

NATARAJAN SELVARAJ

Principal Software Engineer — AI Architect | GenAI & Agentic AI

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SUMMARY

AI Architect with 20+ years of software engineering experience, currently driving enterprise transformation at Autodesk through Generative AI, Agentic AI, and cloud technologies. Specializes in designing multi-agent systems, LLM integrations, and enterprise-scale AI platforms using AWS Bedrock AgentCore, Azure OpenAI, and CrewAI. Proven track record building AI-powered Virtual Assistants, Enterprise Search, and intelligent automation solutions that enhance global workforce productivity.

CORE COMPETENCIES

Agentic AI Architecture · Generative AI (GenAI) · Multi-Agent Systems (CrewAI) · Model Context Protocol (MCP) AWS Bedrock AgentCore · Azure AI Foundry · Azure AI Search · RAG & Vector Search · Microsoft Copilot Studio LLM Integration & Evaluation · LangGraph · Vercel AI SDK · FastMCP · OpenTelemetry · Python · Java · Node.js Serverless Framework · AWS Lambda · CloudFormation · Terraform · Docker · CI/CD · DevOps · Agile

EXPERIENCE

Principal Software Engineer — AI Architect | Autodesk · Novi, MI (Hybrid) | Nov 2024 – Present

- Architected Lumi, an enterprise employee assistant platform with 11+ specialized MCP agents on AWS Bedrock AgentCore, serving Autodesk’s global workforce.
- Built EVA-Crew, a multi-agent AI system using CrewAI with Azure OpenAI (GPT-4o), orchestrating employee queries, automation workflows, and knowledge retrieval.
- Engineered ChatGPT Enterprise ↔ Microsoft Copilot Studio integration via Direct Line APIs, extending coverage to Slack and Microsoft Teams channels.
- Led Kore.AI → custom agentic solution migration using Model Context Protocol (MCP) and AWS Bedrock AgentCore, eliminating platform dependency.
- Implemented enterprise knowledge retrieval with Retrieval-Augmented Generation (RAG) using Azure AI Search and AWS Kendra across multiple knowledge domains.
- Built EA Asset Management MCP server with dual OAuth boundaries (Azure AD inbound / AMIX APIs outbound) and full OpenTelemetry instrumentation for production observability.
- Established AI quality benchmarking using RAGAS evaluation framework and Generative AI Batch Test Framework.

Staff Software Engineer | Ford Motor Company (DeChen) · Dearborn, MI | May 2024 – Nov 2024

- Architected SNAP and SPR Dashboards using Elastic Search, GCP CloudRun, Python, Spring Boot, and Angular UI — reducing page load times from 5–10 minutes to under 1 minute.
- Designed API-first integration layer enabling cross-team data consumption across the GDIA group.

Software Development Manager | Ansira · Remote, MI | Apr 2020 – Mar 2024

- Led engineering delivery and AWS cloud architecture for automotive digital marketing platforms serving 200K+ dealer websites across GM, Ford, Mazda, Nissan, and BMW brands.
- Managed and mentored a 100% DevOps Agile team; defined sprint priorities, conducted code reviews, and drove technical roadmap execution.
- Architected Incentive Imports Framework, reducing new OEM integration development from months to weeks; pioneered Natural Language Search Service for vehicle incentives.
- Drove Oracle → AWS Aurora (Postgres) migration and Redis distributed cache adoption, reducing API response times by 66% (300 ms → 100 ms).
- Built full observability stack: OpenTelemetry, Prometheus, Grafana dashboards, and OpsGenie alerting; introduced Blazemeter load testing for all critical services.
- Containerized 31 Java/Spring applications (Batch + Boot), significantly reducing infrastructure costs and improving deployment portability.

Software Engineer III & IV — Ansira (aka CDK, Sincro), Detroit | Nov 2013 – Apr 2020

- Developed and maintained core platform services including vehicle inventory, offer management, and dealer website integrations for GM, Mazda, Nissan, BMW, and Ford.
- Introduced SonarQube quality gates, Correlation-ID end-to-end tracing, and cross-team knowledge-sharing sessions to elevate code quality and team culture.

Information Architect — General Motors (Mphasis) | Sep 2006 – Nov 2013

- Designed and developed GM Purchasing Web Applications including Global Quality Tracking System, Tool Move, Capacity Planning, and Ferrous & Non-Ferrous Contracting Systems.
- Led Linux environment migration for GM purchasing systems, delivering significant annual infrastructure cost savings.

Software Developer — KAIZEN & LT Infotech India, Chennai | May 2003 – Sep 2006

EDUCATION & CERTIFICATIONS

Master of Computer Applications · CrewAI Certified · Microsoft Copilot Studio Agent Certified · AWS Certified Cloud Practitioner · Sun Certified Java Developer